Data Transformation and Data Preparation in Power Query

Power Query, a core component of Microsoft Power BI, is a powerful ETL (Extract, Transform, Load) engine used for data preparation and transformation before modeling and visualization. It provides a user-friendly, step-driven interface that allows analysts to clean, enrich, and reshape data with minimal coding, while every transformation is automatically recorded as an M query.

This document provides an in-depth explanation of several important data transformation and preparation features available in Power Query, illustrated through a dataset containing client information, including personal details such as names, date of birth, email IDs, and gender. The explanations correspond to operations performed on this dataset, covering column reordering, date transformations, text extraction, delimiter-based transformations, null and blank value handling, and final application of query steps.

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1. Reordering Columns in Power Query

Column arrangement plays a crucial role in making datasets readable, structured, and intuitive for end users. Power Query allows column reordering using two principal methods:

1.1 Reordering by Drag-and-Drop

The simplest method is manually dragging a column header and dropping it into the desired position.

- Select the column.
- Click and hold the column header.
- Drag it to the desired location.

When a column is moved, Power Query automatically records a new step called **Reordered Columns**, ensuring transformations remain reproducible.

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1.2 Reordering Using the Move Commands

Alternatively, Power Query provides structured commands in the Ribbon:

1. Select the column.
2. Open the **Transform** tab.
3. Click the **Move** button.

The Move menu provides:

- **Move Left** – moves the column one position to the left.
- **Move Right** – moves the column one position to the right.
- **Move to Beginning** – shifts the column to the first position.
- **Move to End** – shifts the column to the last position.

For example, if the newly created **Full Name** column appears at the end by default, it can be moved to the beginning using **Move → To Beginning**, and then placed after the **Client Code** column depending on reporting requirements.

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2. Date Transformations: Extracting Month, Year, and Age

Power Query includes robust date transformation functions that enable analysts to extract date components or compute new values such as age, duration, or period information.

For a column like **Date of Birth (DOB)**, Power Query allows:

- Extracting month name
- Extracting year
- Computing age
- Identifying quarter, week, day
- Extracting month start or end dates

- Determining number of days in a month

All these are achieved through the **Add Column → Date** features.

2.1 Extracting the Month Name

To extract the month name from Date of Birth:

1. Select the **DOB** column.
2. Go to **Add Column → Date → Month → Name of Month**.

This creates a new column (e.g., **Month Name**) showing:

- January
- February
- …

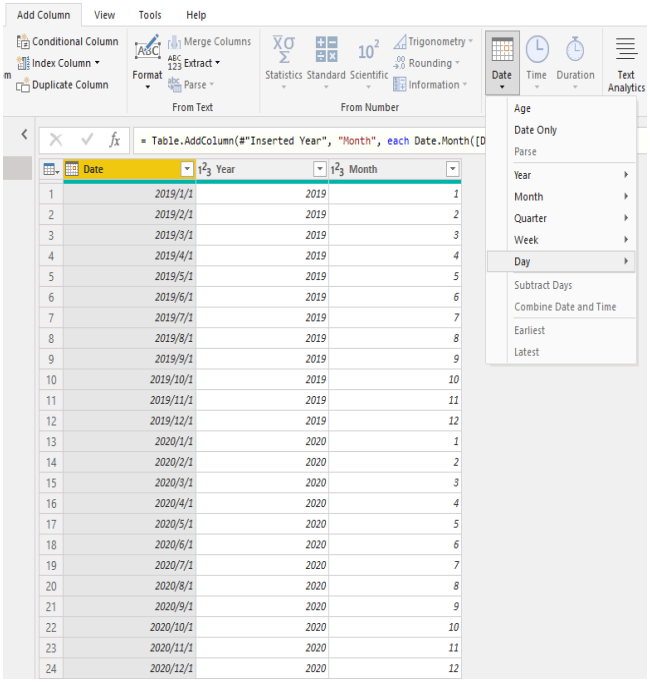
based on each client's birth month.

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Power Query also offers:

- **Start of Month**
- **End of Month**
- **Number of Days in Month**

Each produces a new column reflecting the corresponding dates or numeric values.



2.2 Extracting the Year of Birth

To extract the year:

1. Select the **DOB** column.
2. Navigate: **Add Column → Date → Year → Year**.

This produces a column containing only the year, as a whole number (data type: 123).

Similar options include:

- **Start of Year**
- **End of Year**

These can be useful for more complex time intelligence.

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2.3 Computing Age from Date of Birth

Computing age involves several steps:

Step 1: Use Built-in Age Function

1. Select the DOB column.
2. Go to **Add Column → Date → Age**.

This creates a duration column indicating the total elapsed time between the date of birth and today's date. However, the default output displays **duration in days**, which appears as a long decimal when converted.

Step 2: Convert Duration to Total Years

To transform days into years:

1. Right-click the new **Age** column.
2. Choose **Transform → Total Years**.

The values now show age in years, though still in decimal form (e.g., 59.56).

Step 3: Rounding or Converting to Whole Number

Depending on the reporting requirement, you may:

- Convert to **whole number** (e.g., 60)
  or
- Round to **two decimal places** to preserve precision.

To round:

1. Right-click the Age column.
2. Click **Transform → Round → Round to…**.
3. Specify number of decimal places (e.g., 2).

Thus:

- 59.5589 becomes **59.56**
- 60.4623 becomes **60.46**

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Whole number conversion can also be applied directly by changing the data type to **Whole Number**.

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3. Text Extraction Using Delimiters (Email Username & Domain)

A common requirement is to extract parts of a text string using delimiters.
For email addresses (e.g., *username@gmail.com*), the delimiter is **@**.

Power Query offers specialized text extraction features under:

- **Add Column → Extract**
- **Transform → Extract**

3.1 Extracting Username (Text Before Delimiter)

To extract the username:

1. Select the **Email ID** column.
2. Go to **Add Column → Extract → Text Before Delimiter**.
3. Enter the delimiter: @.

Power Query creates a new column containing everything before the "@" symbol.

Example:

Email	Extracted Username
maria@gmail.com	maria
john.sam@yahoo.com	john.sam

After extraction, rename the column appropriately (e.g., *Username*).

If necessary, convert the text to lowercase:

1. Select the column.
2. Go to **Transform → Format → Lowercase**.

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3.2 Extracting Domain (Text After Delimiter)

Similarly, domain names can be extracted:

1. Select the Email column.
2. Go to **Add Column → Extract → Text After Delimiter**.
3. Set the delimiter to @.

This yields:

Email	Domain
maria@gmail.com	gmail.com
john.sam@yahoo.com	yahoo.com

This helps identify email service providers or validate corporate email structures.

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4. Working with Null and Blank Values

Datasets often contain inconsistent missing values, including:

- cells containing the literal value **null**
- empty (blank) cells

These inconsistencies may distort analysis, produce incorrect aggregations, or lead to invalid data models. Power Query offers structured tools to replace or standardize them.

4.1 Identifying Missing Values

The **Column Quality**, **Column Distribution**, and **Column Profile** features give:

- Percentage of valid values
- Count of errors

- Count of empty cells
- Distinct value distribution

For example:

- Gender column might show:
    - Some rows containing the literal null
    - Some completely blank cells

4.2 Replacing Null Values

To replace null with a meaningful label:

1. Select the Gender column.
2. Go to **Home → Replace Values**.
3. Set:
    - **Value To Find:** null
    - **Replace With:** Others

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This updates all null entries to "Others."

Scenario example:

- Null entries may indicate “Other” gender or non-selection.
  Thus, replacing them improves clarity.

4.3 Replacing Blank (Empty) Values

Empty cells represent true missing entries, not intentional nulls.

To replace blank cells:

1. Select the column.
2. Use **Replace Values**.
3. Keep the *Find Value* field empty.
4. Set the *Replace With* value to **Missing** or **Not Known**.

This ensures all entries are standardized for reporting or visualization.

4.4 When Not to Replace Empty Values

In certain fields—for example, **Prefix** (Mr., Mrs., Ms., etc.)—blank values may be preferable to placeholders like "Missing" because they affect concatenated fields like Full Name.

For example:
If Prefix = Missing
Full Name = Missing John Edward
This appears semantically incorrect.

Therefore, blanks may be left untouched depending on context.

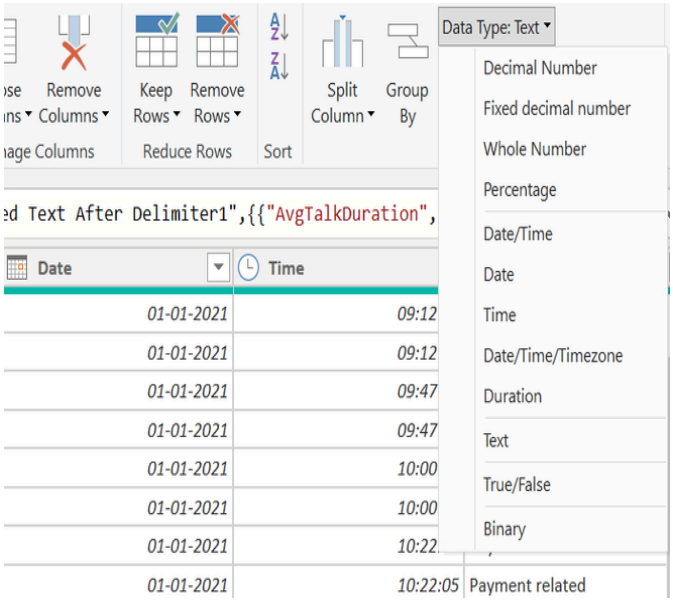
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5. Data Type Considerations

Power Query uses icons to indicate data types:

- **Text (ABC)**
- **Whole number (123)**
- **Decimal number (1.2)**
- **Date (Calendar icon)**



Ensuring correct data types is essential because:

- Sorting behaves differently across types
- DAX measures depend on proper typing

- Visuals interpret numbers/text differently

Example:

- Year column → Whole number
- Month name → Text
- Age → Whole number or decimal
- Domain & Username → Text

Setting the correct type ensures clean modeling.

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6. Using Add Column vs Transform Tab

Understanding the distinction matters:

6.1 Add Column

- Creates a **new column**
- Preserves original data
- Suitable when adding supplemental derived attributes
  (e.g., Month Name, Username, Domain)

6.2 Transform

- Modifies the selected column directly
- Does **not** add a new column
- Used when original column must be altered
  (e.g., lowercase conversion, trimming spaces, replacing values)

Understanding when to use each ensures clarity and prevents unnecessary growth of table width.

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7. Validating Data with Column Quality Tools

Power Query provides built-in diagnostic tools:

7.1 Column Quality

Shows:

- Valid %
- Error %
- Empty %

7.2 Column Distribution

Provides:

- Distinct count
- Unique count
- Frequency distribution

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7.3 Column Profile

Gives statistical summary:

- Min / Max
- Average / Standard deviation (for numeric data)
- Value distribution for text fields

These tools ensure data correctness before loading into the model.

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8. Finalizing Transformations: Close & Apply

After all transformations:

1. Click **Close & Apply** in the Home tab.
2. Power Query:
    - Processes the applied steps
    - Refreshes the model schema
    - Loads transformed data into Power BI

Example:

- Data file size: 744 KB
- Schema sync happens
- Data becomes available for modeling and visualization

Regular saving is recommended to prevent loss during lengthy transformations.

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9.Conclusion

Power Query offers a comprehensive and intuitive environment for performing data preparation tasks before analysis in Power BI. The transformations explored in this lesson illustrate the depth of Power Query's capabilities and their real-world applications:

- Column reordering improves readability and usability.
- Date transformations extract meaningful time components for analysis.
- Age calculation demonstrates how duration and transformation functions work.
- Text manipulation using delimiters is vital for parsing email IDs or structured text.
- Handling null and blank values ensures data integrity.
- Choosing between Add Column and Transform affects both structure and logic.
- Column profiling provides visibility into data quality.
- Finally, closing and applying the query loads clean, well-prepared data for visualization.

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Such systematic preparation ensures that downstream processes—DAX modeling, visualization creation, and reporting—are built upon accurate, consistent, and analysis-ready data. By mastering these Power Query features, users can transform raw datasets into meaningful, business-ready datasets that enhance decision-making and deliver actionable insights.